1 Introduction

1. Definition of masting
    (a) Observations of masting in relation to seed predation and flowering
2. Different hypotheses operating at different reproductive stages like filters
    (a) First, pollination success (Pollination coupling)
    (b) Second, resources availability during development phase (Resource Matching)
    (c) Third, dispersal (Predation satiation)
3. Why traits can help link hypotheses to masting mechanisms
    (a) Mechanisms are not mutually exclusive, but some might act stronger than the other for certain species
    (b) Traits thus can be important to understand masting through their relative importance
4. Trait-based perspective
    (a) Literature review on understanding masting from traits perspective
        i. Wind pollination is related to masting
        ii. Animal dispersed seeds have higher masting than wind dispersed seeds
        iii. Larger seed is associate with masting
        iv. All these studies use CV to quantify masting but we use a binary definition, whether this species is a strong masting species or weak masting species on a broader species-level
        v. If evolutionary associations are sufficiently strong, we should be able to detect it even under a coarse binary classification
        vi. Recent papers also focus on resource perspective using leaf and shoot traits
    (b) Existing studies mainly focus on ecological mechanisms.
    (c) Next step may be clear connection between traits and hypotheses and potential evolutionary drivers for certain species
5. Introduce framework linking hypothesis x traits x masting
    (a) Linking traits to specific hypotheses helps us understand the underlying drivers of masting across different species
    (b) This framework is especially useful when considering whether evolutionary drivers are more similar within closely related species (clades)
6. We try to do this using data from one specific geographical region

(a) Here we use North American tree data to test this framework since they have information on their life-history
    i. Masting info available
    ii. Includes both strong masting and weak masting species
    iii. Not using CV data because don't have stand-level traits data

2 Discussion

1. Summary of main findings
    (a) Pollination supports the pollination coupling hypothesis
    (b) Seed weight supports the predation satiation hypothesis
    (c) These act at the beginning and end of the reproductive cycle, two critical stages for successful reproduction and gene transmission
    (d) Discuss mechanisms in the context of existing literature
        i. Previous literature on seed mass and masting think from the resource depletion perspective (Bigger seeds use up resources so cannot reproduce every year)
        ii. However, from evolutionary perspective, seeds attractive to animal dispersers will benefit more from masting
        iii. At the same time, they have to make sure the seed survived can established
        iv. Selection may favor coordinated strategies that increase seed survival across the recruitment stages (Bonal *et al.* (2007); Moles & Westoby (2004))
        v. Relationship between masting and seed weight is stronger within biotic-dispersed group
    (e) Other traits show some pattern but statistically non-significant
    (f) Transition to alternative hypotheses
2. Masting is a population-level phenomenon
    (a) Other traits may matter to masting below the species level, like population level
    (b) Examples of species that mast only under certain environmental conditions (E.g., *Hymenaea coubaril*)
    (c) To address, this traits need to be measured locally and match with masting data – we need more data below the level of species!
3. Different species selected under different hypotheses
    (a) Introduce flowering masting vs. seed masting
    (b) Different species driven by different mechanisms

    (c) Need a more refined definition of masting and additional flowering-related traits
    (d) Transition to phylogeny

4. Phylogenetic signal for traits and masting
    (a) Many traits have strong phylogenetic signal, it is hard to tease out traits from masting, which limits the inference
    (b) The lit on this mostly talk about adaptive significance, we need to understand its broader macroevolutionary patterns
    (c) Data is strongly nested in phylogeny, and data is sparse.

5. Deep evolutionary split between angiosperm and gymnosperm
    (a) Gymnosperms and angiosperms occupy fundamentally different functional spaces
        i. More masting gymnosperm sp
        ii. Longer leaf longevity in gymnosperm
        iii. More monoecious species in gymnosperm
        iv. Wind pollination is universal among gymnosperms
        v. Importance of looking for potential evolution support between gymnosperm and angiosperm
            A. Masting is more common in gymnosperm with specific functional traits
            B. Test if masting was gradually diluted in angiosperm as they transitioned to biotic pollination and faster life histories

6. Importance of testing within clade
    (a) Existing literature points out the potentials within certain clade for evolutionary investigations (Satake & Kelly (2021))
    (b) This requires more data collection on species-level and population-level within certain genus groups, for example: (*Quercus.*)
    (c) So we need more data lower than species and more targeted data across species that considers deeper evolutionary time (we're missing both the super recent evolution. – populations – and the super deep)

References

Bonal, R., Munoz, A. & Díaz, M. (2007) Satiation of predispersal seed predators: the importance of considering both plant and seed levels. *Evolutionary Ecology* **21**, 367–380.

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